# Chen-MA-analyses

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The following analyses should be done for each simulation condition (in decreasing order of priority of conditions):

- collaborative task with full prompts
- collaborative task without detailed character profile
- [baseline 1] case solving task with default zero-shot LLM prompt (as you already did it to verify that the cases are solvable)
- [baseline 2] case solving task with character prompts, but without collaboration, to the LLM (as discussed during the meeting)
- collaborative task without detailed rules

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- ~~debate task with full prompts~~
- ~~debate task without detailed character profile~~
- ~~[baseline 1] debate (maybe framed as an argumentation?) task with default zero-shot LLM prompt~~
- ~~[baseline 2] debate (maybe framed as an argumentation?) task with character prompts, but without interactive competition, to the LLM (as discussed during the meeting)~~
- ~~debate task without detailed rules~~

1. Descriptive analyses
    a. Calculate average task accuracy & CIs in the different conditions
        i. if you actually find lower performance, you could frame this result closer to your original research questions — as investigating whether it is breakdown in character consistency that leads to the lower performance. If evaluation results will not support this, you can just

say that there might be other reasons for the lower accuracy, to be explored in future work ;)

ii. task success is straightforward yo define for the collaboration task, but how will you measure it for the debate task? Please make a suggestion :)

b. Look at the average nr of turns, avg. length of single turns (e.g. number of words)

2. Classifier model
    a. BERT based text classification model as a model-based baseline to evaluate character consistency.
        i. will be trained on dialogue utterances from original Sherlock Holmes, Hercule Poirot, and Miss Marple texts (all in the public domain), mapping input text to one of four classes: SH, HP, MM, or Other
        ii. ‘Other’ should probably contain dialogues from other characters in the same books to match other properties of the distribution
        iii. Will provide us reference texts, which we can then also use for other measures
    b. Test accuracy of the classifier on a test split
    c. Assess the classification confidence development over dialogues
        i. Extract the predicted probability of the correct class and plot mean (over simulations) over turns, CIs over the simulations
        ii. Calculate average (over simulations) Brier score over turns (mean squared error of predicted probabilities)
        iii. Fit linear regressions to the scores over turns, in order to estimate whether there is a significant increase or decrease trend
3. Lexical measures
    a. Character-specific vocabulary rate
        i. Extract TF-IDF vectors from the gold standard dialogues of each character (avg) and the single turns of each character, and  calculate cosine similarity between the tf-idf vectors

        ii. Alternatively (only if tf-idf will be infeasible): Calculate type-token ratio of the ground truth dialogues and the turns of each character (i think this doesn't quite capture character specifics of vocabulary, though)

    b. Intra-agent cosine distance

        i. Baseline for distinguishing similarity that just comes from similarity of turns on the same topic from in-character similarity:

            1. character distance = (cosine similarity between turns) - (cosine similarity between current turn & turn at same index from a different character)

        ii. note: to be interpreted with caution, strongly depends on the embedding model

    c. ~~Vocabulary contamination marker → let's turn this approach into cluster separability analysis~~

        i. ~~original proposal: keyword hit rate, but since it depends on manual specification of relevant keywords, and chances are that the hit rate will just be 0, I suggest to analyze contamination through clustering and the separability of clusters, as explained under "clustering" below.~~

4. Syntactic level

    a. Dependency parsing with Spacy of

        i. a sample (e.g., 100) reference utterances in the corpus constructed for fine-tuning the classifier for each character (reference parses)

        ii. sentences from turns from each character, e.g., from 2 simulations (simulation parses)

    b. Calculate average and SD of the maximal dependency tree depth of the reference and simulation parses

    c. Calculate the difference between reference and depths for each turn, fit linear regression model over turns (to see if there is a significant trend for increasing or decreasing differences in syntactic complexity relative to the reference texts)

5. Discourse Level

a. Dialogue act markers: labels sentences in each turn with communicative intent:

    i. use a pretrained model, e.g.: https://huggingface.co/Falconsai/intent_classification (not sure which set of labels they use exactly) to label the types of utterances (single sentences?) in the simulation turns, and in the corpus used for fine-tuning the classifier

    ii. calculate the proportions of different types of intents, for each character, in the turns and in the reference corpus, and calculate a KL divergence between those proportion distributions

    iii. this will be more of a descriptive analysis, I think, since we will not be looking at any trends

b. Sentiment trend: Measures emotional trajectory of each character —- Methods: VADER

    i. calculate the average positive / negative / neutral sentiment scores over a sample of sentences from the corpus of sentences used for the classifier training for each character (reference)

    ii. calculate the sentiment scores for sentences from turns of each character, average within turn

    iii. calculate e.g. the euclidean distance between the score vector of a turn and the reference score vector, and fit a linear regression to see if the distance increases or decreases over turns

c. ~~Stance marker: i would recommend not doing this analysis~~

d. Detective methodology contamination marker: mark when character adopts another's methodology. — Methods: Manually tagging or rule-based

    i. I think this analysis makes more sense for the debate task, not the collaboration task?

    ii. This seems to be indeed only possible with manual annotation; therefore, I would like to ask you to do this as the very last analysis, only if there is time once everything else is finalized.

e. Discourse pattern deviation marker: Checks internal contradictions in dialogue which violate its original discourse pattern. — Methods: Manually tagging or rule-based

    i. I am not sure what exactly was meant here. I would also suggest to not do this part of the analysis.

6. Validate the measures above via comparing all the measures above against the performance of the classifier

    a. we would like to see if the measures we proposed above, since they are new for analyzing character consistency, actually capture what they are supposed to capture. To that end, we treat the predictions of the classifier at each turn (i.e., the probability of predicting the true character) as ground truth, and compare *each* (turn-wise) metric above against it

        i. we plot each pair and calculate a correlation (1 would be ideal, 0 would mean that our new measures capture noise or something different)

            1. this will allow us to identify the most "faithful" linguistic metric

        ii. calculate an expected calibration error between the classifier and each linguistic measure

7. Clustering (as a measure of contamination between characters)

    a. create vectorized representations of each turn (from the simulation), and each utterance from the reference corpus used for classifier training

        i. approach 1: embed them with the same model as you use for the cosine similarity analysis → apply dimensionality reduction

        ii. approach 2: create a vector concatenating all the turn-wise metrics above

    b. cluster the vectors into 3 clusters (corresponding to each character), e.g. via k-means

        i. cluster the simulation turns separately from the reference corpus turns

    c. evaluate both clustering results to check for their separability between character clusters

        i. calculate Silhouette Coefficient

ii. calculate purity of the clusters

iii. compare the results between the simulations and reference corpus, and for each type of the vector representations

8. Analyses focusing on comparing the turns vs initial / final reports

    a. if you are still planning on eliciting these, please let me know if there are specific analyses you plan to do on these other than the analyses defined above

→ Based on these analyses, we would like to draw conclusions about :

- the LLMs’ abilities to stay in character
- which linguistic measures are best to capture this
- the LLMs’ performance in such a complex task, and what prompting worked best (we will be able to tell this based on analysis results of the different conditions)
- [ i know that these are slight paraphrases of your original research questions, but I think these are most operationalizable]